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Module H – Cybersecurity and National Defence

L1 - Cryptographic Sovereignty: Secrecy, Intelligence and State Power



**Politecnico
di Torino**

Department of Control and
Computer Engineering



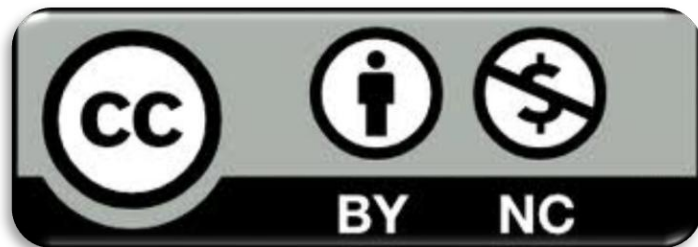
TORSEC

Computer and Network Security Group

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- This material is used and adapted within the course “Cybersecurity and National Defense”.



If a system is secure publicly, is it secure against a state?

Public security $\neq$ security against state-level capabilities

Cryptography as national capability is divided into three aspects:

- **Defensive capability:** Protect state, military, diplomatic, and industrial secrets.
- **Offensive/intelligence capability:** Read, alter, or exploit other people's secrets.
- **Sovereign capability:** Do not blindly depend on uncontrollable algorithms, products, standards, hardware, firmware, or suppliers.



Learning objectives

By the end of this lecture, you should be able to explain:

1. Why cryptography is a national defence capability
2. How cryptanalysis becomes strategic intelligence
3. Why secrecy around capabilities matters
4. Why dependence on uncontrollable algorithms, products, hardware, or suppliers compromises part of a state's sovereignty
5. How historical cases anticipate modern cyber defence

Cybsec protects systems. National defence protects state power.

Cybersecurity focus

- Systems
- Data
- Users
- Networks
- Services



National defence focus

- Sovereignty
- Command
- Intelligence
- Military operations
- Critical functions



Cryptography is an evolutionary arms race

The history of cryptography is not linear progress. It is a continuous competition.

- **Code-makers** design systems to protect secrets.
- **Code-breakers** search for weaknesses and exploit them.

When a system is broken, it must evolve, or become useless.

Code-makers

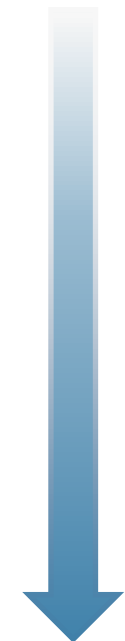
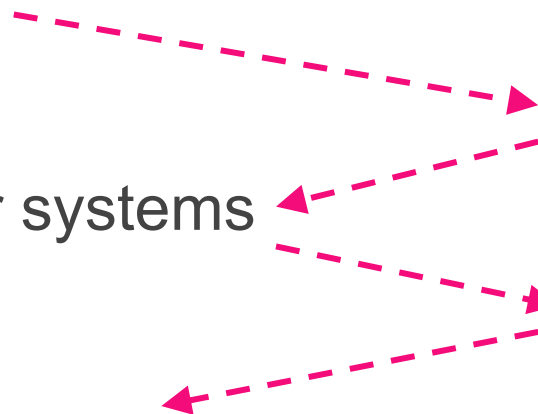
Protect secrets

Design stronger systems

Code-breakers

Break protection

Find new weaknesses





What changes the balance between secrecy and exposure?

The balance between code-makers and code-breakers changes when new capabilities appear:

1. Mathematics

Frequency analysis, probability, number theory, computational theory

2. Machines

Telegraph, radio, rotor machines, computers, quantum computers

3. Organizations

Cryptanalysis is a state function: cipher bureaus, black chambers, intelligence agencies

4. War and political pressure

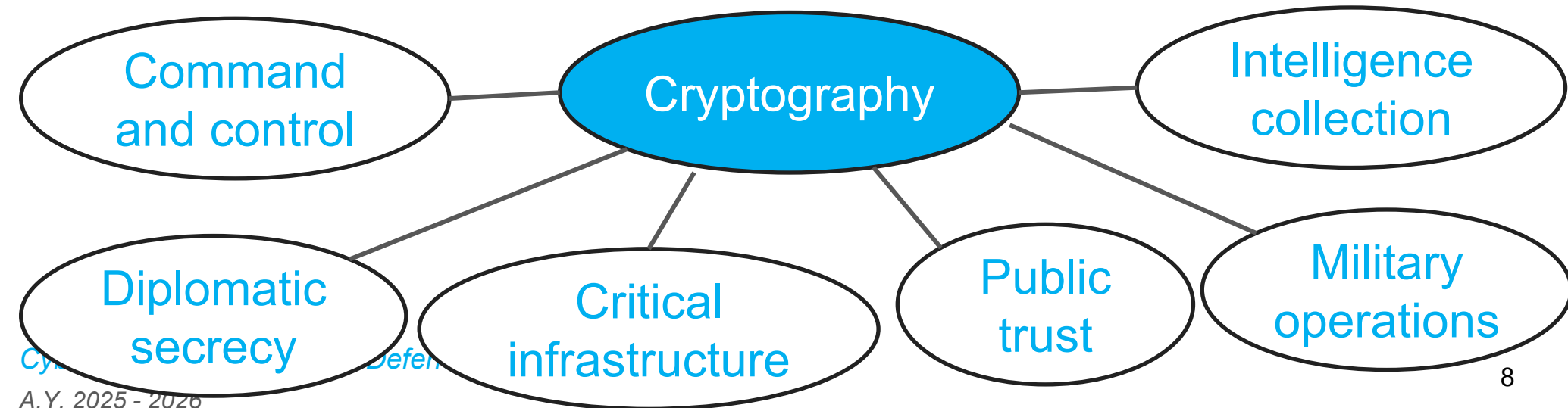
War accelerates innovation and rewards secrecy

Why cryptography becomes national defence

- **Cryptography enables the state to keep secrets at scale.**

Cryptography becomes national defence when it protects: military orders and command chains, diplomatic negotiations, intelligence sources and methods, operational plans, strategic infrastructure, the credibility of state institutions.

- Breaking cryptography can therefore produce **strategic effects**.





Knowing how to break secrecy is itself secret

The paradox of cryptanalysis

If you reveal that you can break a system,
the adversary changes the system.
So the capability **must remain hidden**.



States may be decades ahead of the public domain

Public domain

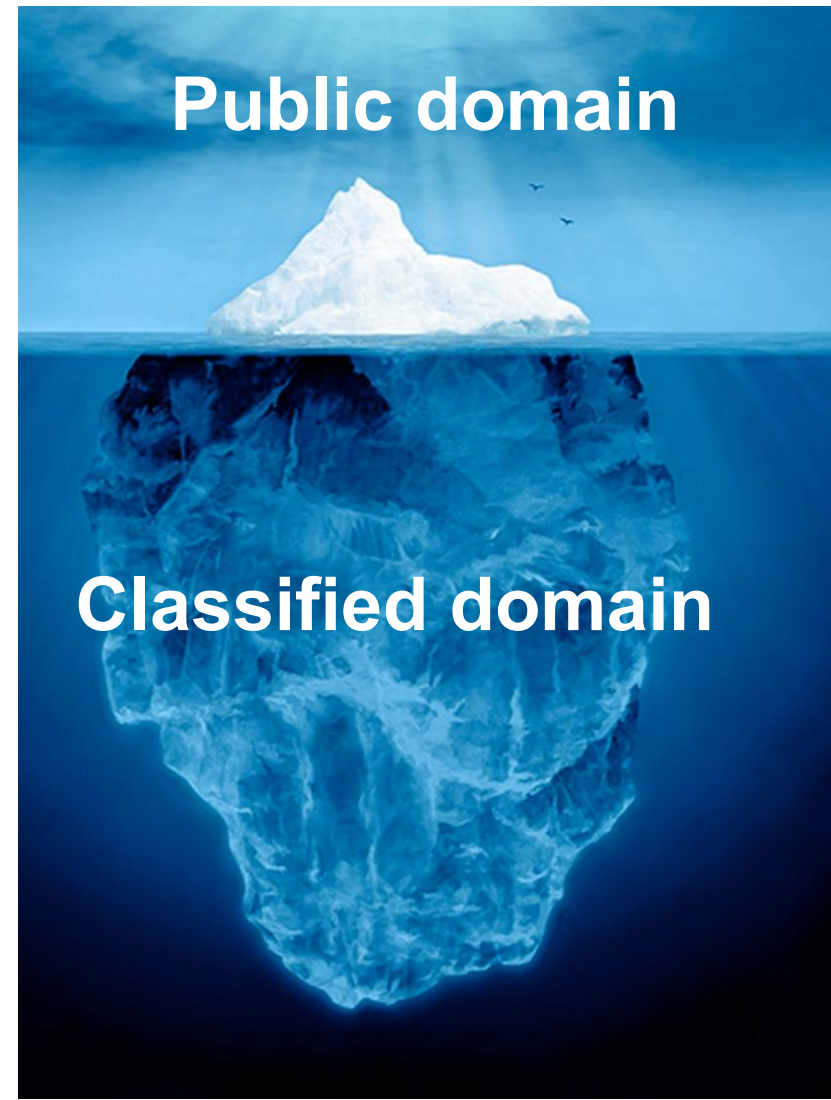
Academic papers, standards,
civil technology

Classified domain

Signals intelligence, military research,
national laboratories

Question

What do States know that the
public does not?





What historical cases teach us about state behaviour

We are not studying history for curiosity.

We are looking for recurring patterns in how states use cryptography and cryptanalysis.

Three questions:

1. How do states exploit weak cryptography?
2. How do states hide cryptanalytic breakthroughs?
3. Why does cryptographic sovereignty matter?

Case 0: Mary Stuart and the Babington Plot

Excessive trust in a cryptographic tool can be risky when very high interests are at stake

Why Mary Stuart was a national security threat

England, late 16th century

Mary Stuart, Queen of Scots, was imprisoned in England and seen as a Catholic rival to Queen Elizabeth I.

Babington Plot, 1586

A group of conspirators planned to:

1. assassinate Elizabeth I;
2. free Mary Stuart;
3. place Mary on the English throne.

The conspirators used encrypted letters to communicate with Mary.



Mary Stuart



Elizabeth I



The communication channel was already compromised

Mary's encrypted correspondence did not travel through a secure channel.

Walsingham's network controlled the route:

- the courier channel was infiltrated;
- letters were hidden and smuggled;
- messages were secretly copied;
- originals were resealed and delivered;
- Mary did not know the channel was compromised.



Encryption does not secure a compromised communication system.



Mary's cipher looked secure ... but it was breakable

Mary's system was a **nomenclator**:

- symbols replaced letters;
- some symbols represented common words or phrases;
- null symbols were added to confuse analysis;
- the system looked sophisticated to its users ...

... but it still leaked structure: it preserved linguistic regularities, repetitions and contextual clues

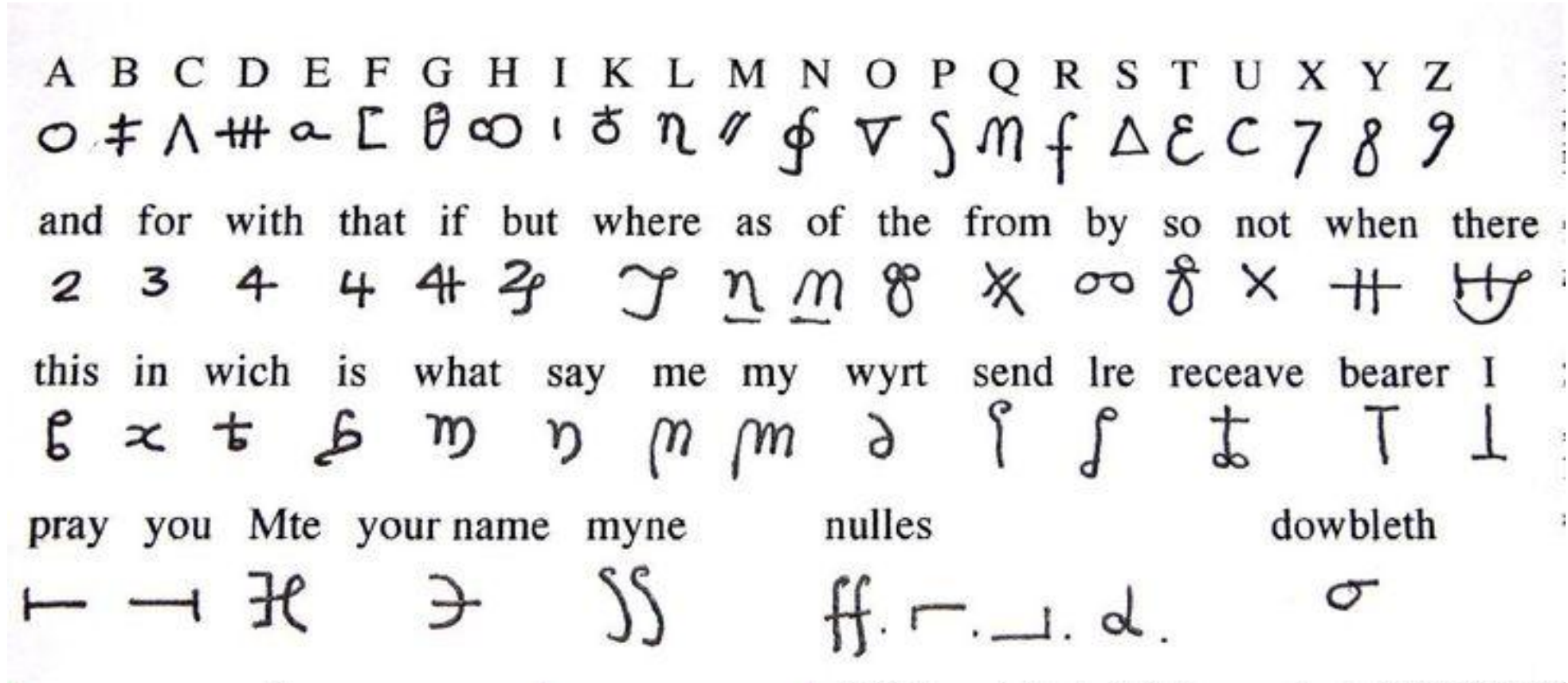
Weakness:

A **skilled cryptanalyst** could exploit linguistic patterns and context.

Thomas Phelippes, working for Walsingham, reconstructed the plaintext



Mary Stuart's nomenclator





From decryption to deception

- Once Walsingham could read the correspondence, he did not immediately stop the plot. He waited because he wanted stronger evidence and the names of the wider network.
 - Forged postscripts were inserted into the correspondence to confirm Mary's involvement and identify additional conspirators.
- The decrypted letters became political and legal evidence.
 - Babington and the conspirators were arrested and executed.
 - Mary Stuart was tried for treason.
 - Mary was executed in 1587.
- The episode removed a Catholic claimant to the English throne and helped protect Elizabeth I's Protestant regime.
- Without this cryptanalytic success, England's religious and political trajectory might have been very different.



Cryptanalysis became an active instrument of state power.



Weak encryption may be worse than no encryption

Mary and Babington trusted the cipher. Because they trusted it, they wrote explicitly. Because they wrote explicitly, the decrypted letters became evidence.

National defence lesson:

- Weak cryptography (or compromised communication channel deemed secure) can change behaviour in ways that help the adversary.
 - **No encryption:** Users may self-censor.
 - **Strong encryption:** Users can communicate securely if implementation and procedures are sound.
 - **Weak encryption believed to be strong:** Users speak openly while the adversary reads silently.



Cryptanalysis works best when combined with intelligence operations

In the Babington Plot, the state did not merely “break a cipher”.

It combined:

- human intelligence
- covert interception
- document copying and resealing
- cryptanalysis
- deception
- legal-political exploitation

The adversary did not only break confidentiality. It also achieved persistence in the communication channel and used the compromised channel for active manipulation.



Secure communication is not only about the algorithm.
It is about the entire communication ecosystem (people, trusted endpoints, procedures, and operational discipline.)

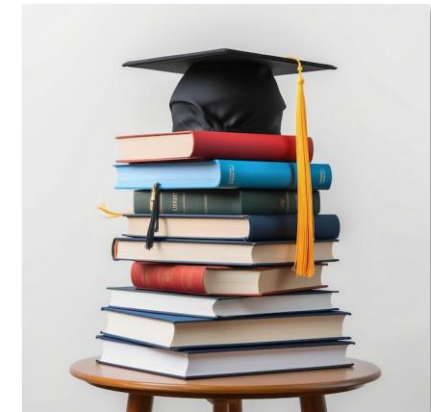
A cryptanalytic breakthrough is a strategic asset

Breakthrough + secrecy = intelligence advantage

Breakthrough + disclosure = adversary adaptation

Once a state can break a system, it faces a choice:

- publish the discovery $\Rightarrow$ not convenient to the interests of the state
- use it secretly, to preserve the advantage and prevent the adversary from adapting



This is where cryptography becomes statecraft.

This creates a structural tension between science and intelligence:

- public science rewards disclosure
- intelligence rewards silence

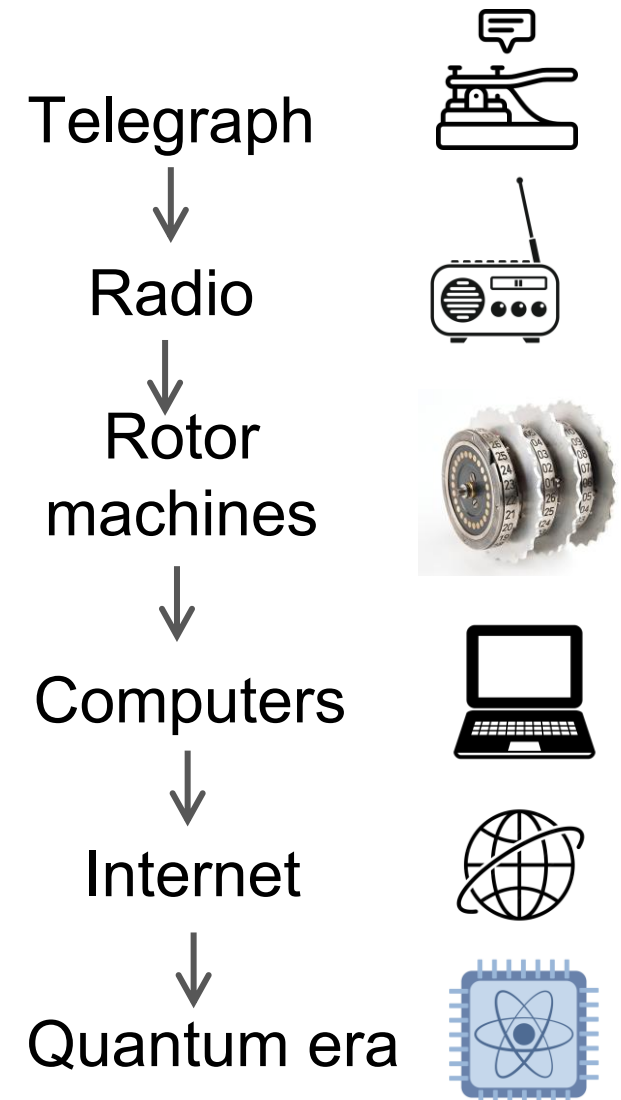


Every communication revolution creates a security crisis

New communication technologies increase:

- speed
- scale
- distance
- dependency
- interception opportunities

The more strategic communication becomes, the more strategic cryptography (and in general, secure communication) becomes.



Case 1: The Zimmermann Telegram

World War I: Britain needed the United States in the war

Europe, early 1917

Britain and the Allies

Wanted the United States to enter the war.

Germany

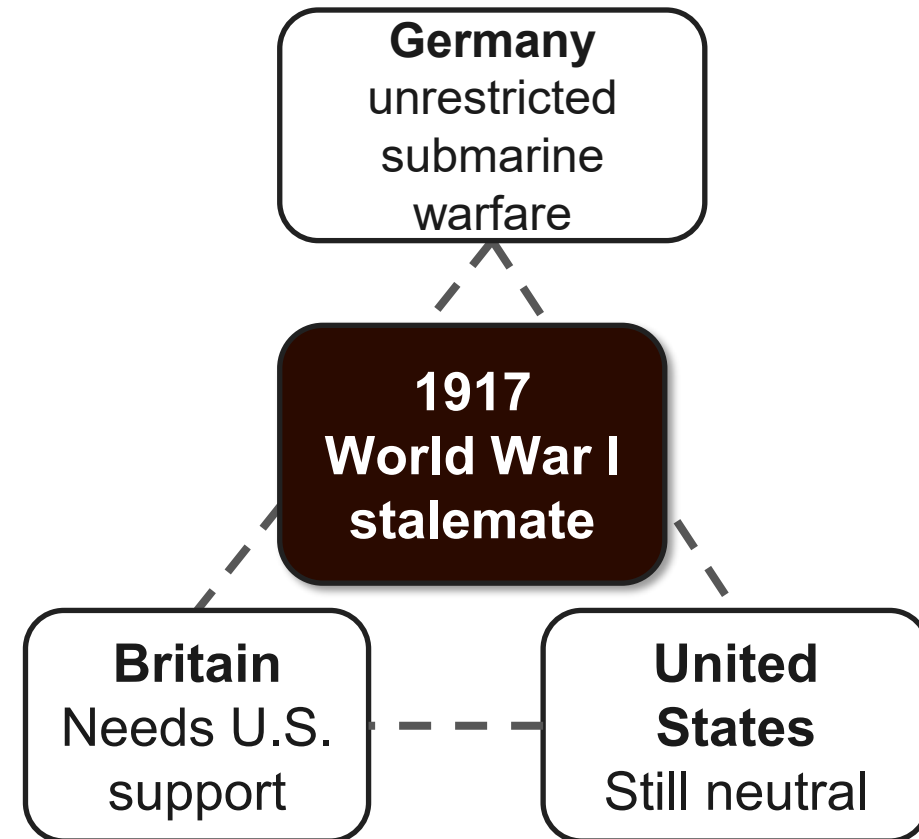
Planned unrestricted submarine warfare to cut British supply lines.

United States

Was still officially neutral, but increasingly under pressure.

Strategic problem

Germany needed to win quickly before U.S. intervention could change the balance.





The **Zimmermann Telegram**: when **cryptanalysis enters grand strategy**

January 1917: German diplomatic message to Mexico

Germany proposed that, if the United States entered the war, Mexico should join Germany against the U.S.

In return, Germany suggested support for Mexico to recover:

- Texas
- New Mexico
- Arizona

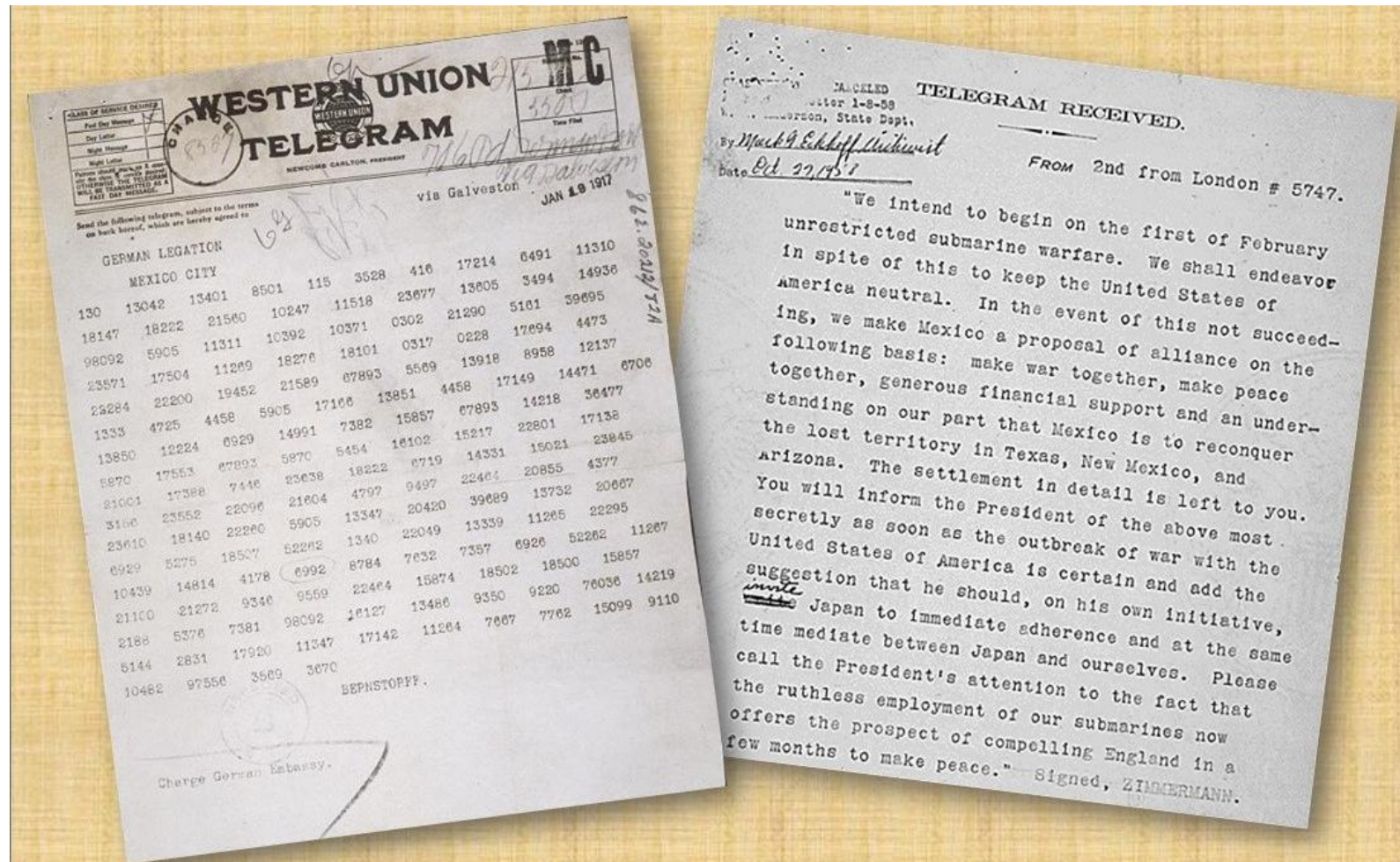
From a German perspective, the logic was strategic distraction: if the United States had to worry about a war on its southern border, it would be much harder for Washington to send forces to Europe quickly.

British intelligence intercepted and decrypted the message.

Its disclosure helped shift U.S. public opinion and policy.



The Zimmermann Telegram decoded

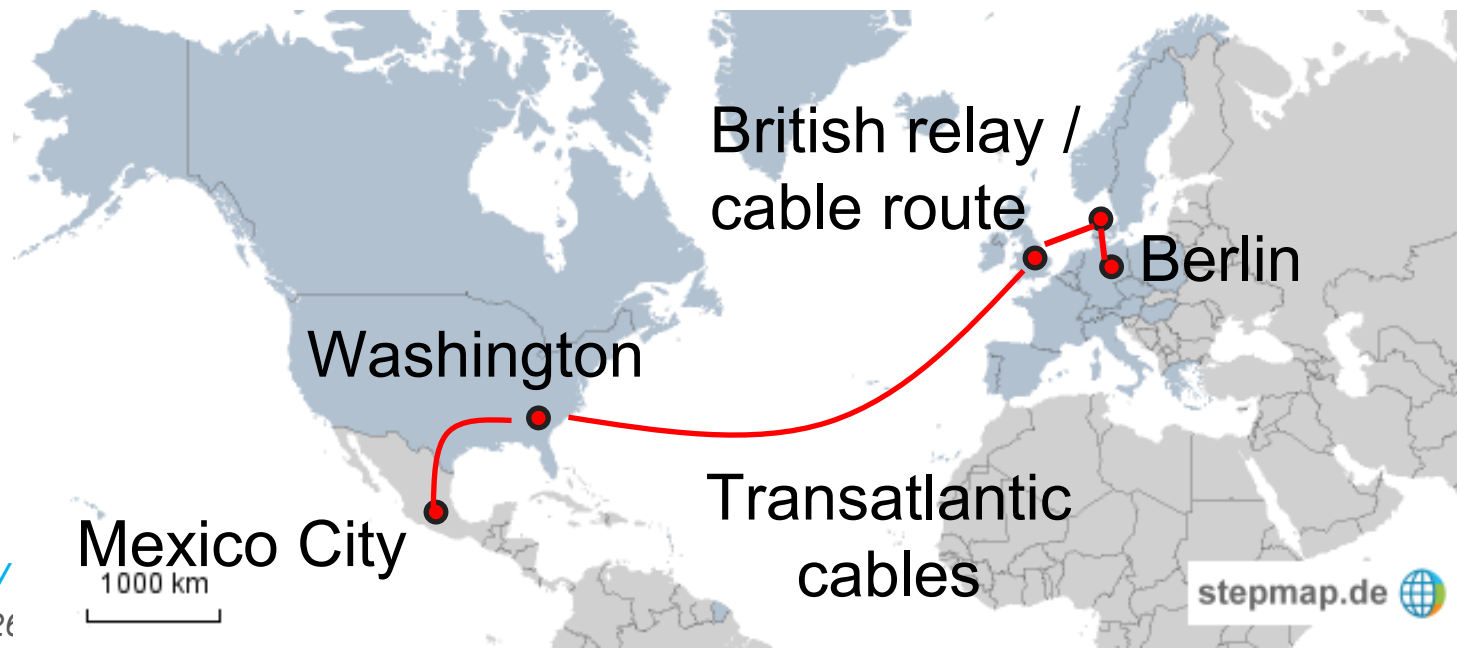


Why the message was vulnerable: encrypted content, exposed infrastructure

- The telegram was vulnerable because diplomatic communication depended on infrastructure the sender did not fully control.

Strategic dependency:

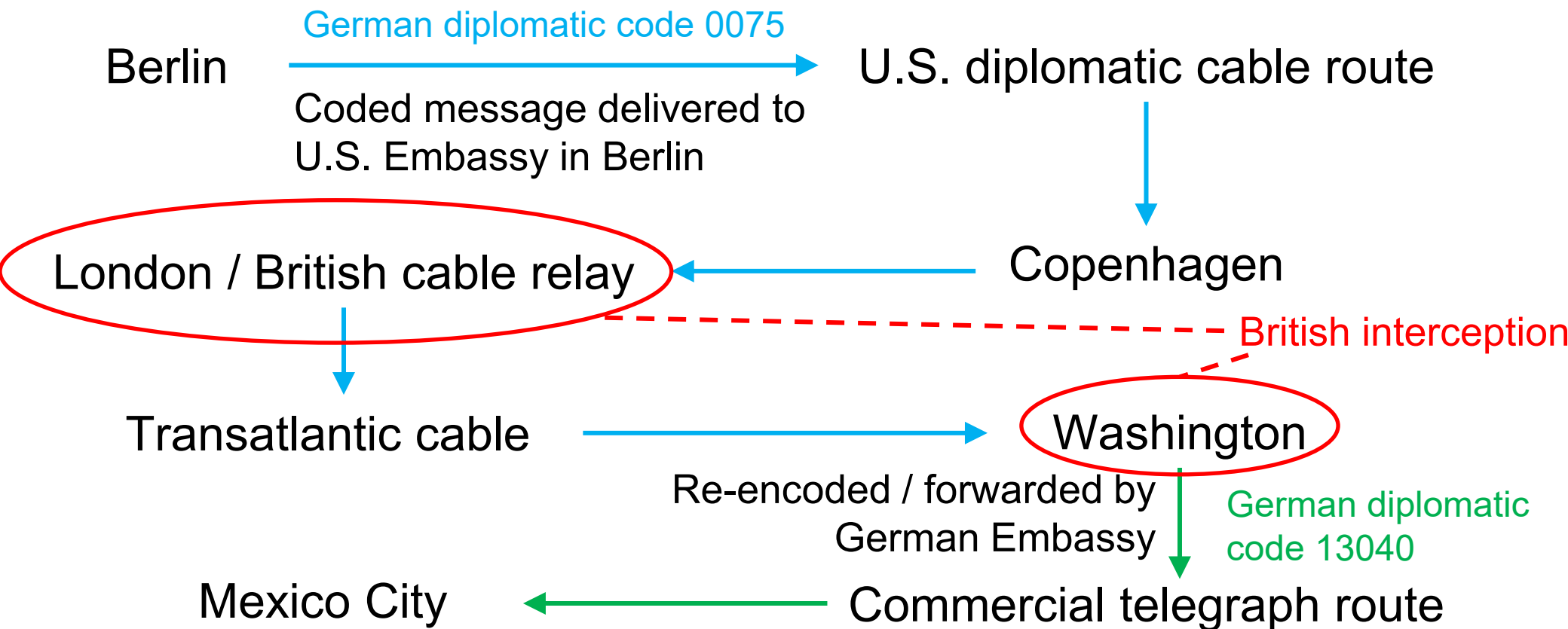
If your communication path crosses adversary-controlled or third-party infrastructure, secrecy depends on more than encryption.





The communication environment

Germany's own Atlantic cables had been cut earlier in the war. To reach Mexico, Germany relied on diplomatic/neutral infrastructure:





Attualità > Politica Internazionale

Gli Stati Uniti stavano spiando la Merkel

 **Giorgia Bonamoneta**  1 Giugno 2021 - 00:05

Angela Merkel, Cancelliera tedesca, è stata spiata dai servizi segreti di due Paesi: da una parte l'NSA americana e dall'altra le forze di intelligence danesi (FE). Altri politici e funzionari coinvolti. Macron e Merkel chiedono spiegazioni immediate.

Germania: **Angela Merkel**, Cancelliera federale della Germania in carica, è stata spiata dal servizio di intelligence americano, lo US National Security Agency (NSA), in collaborazione con le Forze Armate del Servizio di intelligence danese (FE).

La notizia di oggi, 31 maggio 2021, ha colto tutti di sorpresa. La notizia è stata subito aggiornata e così si è scoperto che lo spionaggio non era indirizzato unicamente contro la Cancelliera o gli altri due nomi della politica tedesca citati nell'inchiesta, **Peer Steinbrück** e **Frank-Walter Steinmeier**, ma anche verso tutti i maggiori esponenti della politica europea.





From intercepted signal to usable intelligence

A ciphertext is not yet intelligence. To become intelligence, it needs:

1. Intercepted ciphertext
2. Cryptanalytic work for decryption
3. Partial reconstruction / interpretation
4. Confidence assessment
5. decision-maker trust / political use

Use the intelligence ... or protect the source?

Cryptanalytic intelligence creates a dilemma:

- **If used publicly:** it may shape events.
- **If used too visibly:** it may burn the source.

Should we reveal
the telegram?

Reveal

- Influence U.S. policy
- Risk exposing British codebreaking

Protect

- Preserve access
- Lose strategic opportunity



Cryptographic sovereignty: who controls the tools that protect state secrets?

Cryptographic sovereignty is not only about using encryption.

For national defence, a state must be able to trust and govern:

- which algorithms and standards are used;
- who implements and verifies them;
- how keys are generated, stored, rotated and destroyed;
- which hardware and firmware execute cryptographic operations;
- who owns, updates and maintains the products;
- how operators use the system (do they reuse keys, bypass procedures, use predictable configurations?).

Cryptographic sovereignty is the ability to protect state secrets without blind dependence on actors outside national control.

Case 2: Crypto AG / Operation Rubicon



Crypto AG: when the supplier became the intelligence operation

Crypto AG was a Swiss supplier of encryption machines used by governments for diplomatic, military and intelligence communications.

What happened

- **1950s–1960s:** Crypto AG becomes a trusted global supplier of cipher machines.
- **1970:** CIA and West German BND secretly acquire control of the company.
- **1970s–1990s:** Manipulated products are sold to foreign governments.
- **1993:** BND leaves the operation; CIA reportedly continues.
- **2018:** Crypto AG is dismantled / assets sold.
- **2020:** The operation is publicly exposed by investigative reporting (Washington Post, ZDF and SRF), followed by Swiss investigations.

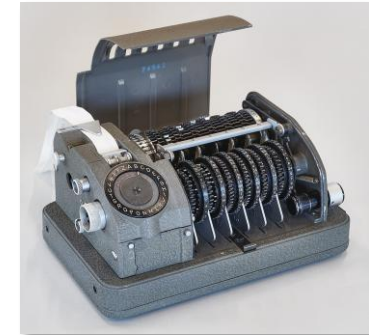
[How the CIA used Crypto AG encryption devices to spy on countries for decades - Washington Post](#)
[The CIA's 'Minerva' Secret | National Security Archive](#)

Governments bought “secure” encryption devices. Some devices were allegedly designed so that U.S. and German intelligence could read the traffic.

The strategic damage scales with the customer base

Reported exposed customers included (the full target list remains partly opaque.):

- Middle East: Iran, Egypt, Saudi Arabia, Libya
- South Asia: India, Pakistan
- Europe: Italy, Vatican
- Latin America: Mexico, Argentina, Brazil, Chile, Uruguay, Peru, Colombia, Venezuela, Nicaragua



The Crypto AG case shows that a supplier-level compromise can scale across states. Why this matters:

- one vendor can serve dozens of governments
- one hidden design weakness can expose many networks
- diplomatic and military communications reveal intentions, alliances and crises.
- a compromised product can operate silently for decades.



In national defence, cryptographic procurement is not only a technical decision. It is a strategic decision about whom the state is willing to trust.

Case 3: Enigma, Bletchley Park and Ultra

When a trusted cryptographic infrastructure
becomes an intelligence source

Enigma was not just a machine: it was military infrastructure

World War II, 1939–1945

Germany relied on Enigma to protect large volumes of military communication:

- army operations, naval movements, and air force coordination;
- operational orders and logistics;
- weather reports;
- submarine warfare;
- coordination of rapid military operations.

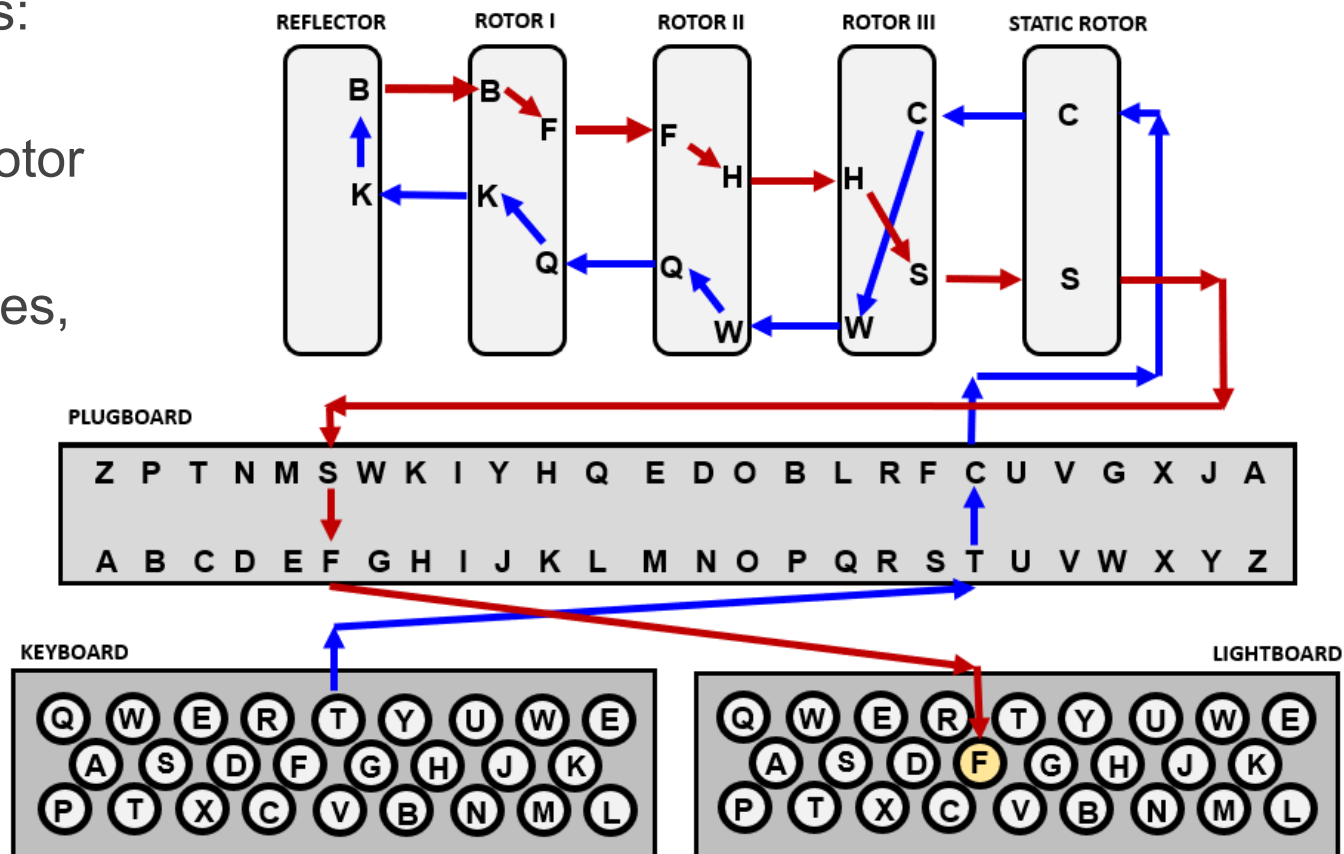


Inside Enigma: one keypress, many transformations

When an operator pressed a key, the signal passed through several layers.

Rotors step after keypress:

- After each keypress, at least the rightmost rotor advanced
 - After enough keypresses, the next rotor moved as well, and so on
- ⇒ **the cipher changes dynamically**





Why Enigma looked secure

Dynamic encryption

The rotors moved as the operator typed.

The same plaintext letter could produce different ciphertext letters.

Combinatorial explosion

Security depended on several settings:

- rotor order;
- rotor starting positions;
- ring settings;
- plugboard connections.

Even simplified configurations produced an enormous search space.

Operational keying

Operators used **key sheets** to configure Enigma, i.e., secret documents distributed in advance to the units:

- **daily key**: shared by the communication network for that day;
- **message key**: chosen for a specific message.

German assumption: Complex machine + huge key space + daily procedures
⇒ Brute force was infeasible, so intercepted traffic would remain unreadable.

Before Bletchley: the Polish breakthrough

The first major breakthrough against military Enigma came from Poland. **Why Poland?**

- Germany adopted Enigma for military communications in the 1920s.
- Britain and France initially failed to break it.
- Poland faced a direct strategic threat from Germany.
- For Poland, German encrypted traffic was a national survival problem.



Marian Rejewski

Polish cryptanalysts treated Enigma not as a language puzzle, but **as a mathematical machine.**

Marian Rejewski, together with Jerzy Różycki and Henryk Zygalski, brought mathematics into the centre of cryptanalysis.



How Poland started to break Enigma

The Polish breakthrough combined four elements:

1. **Intelligence input:** French intelligence obtained German Enigma material from Hans-Thilo Schmidt and shared it with Poland.
2. **Procedural weakness:** German operators encrypted the message key twice.

Example:

Message key: P G H

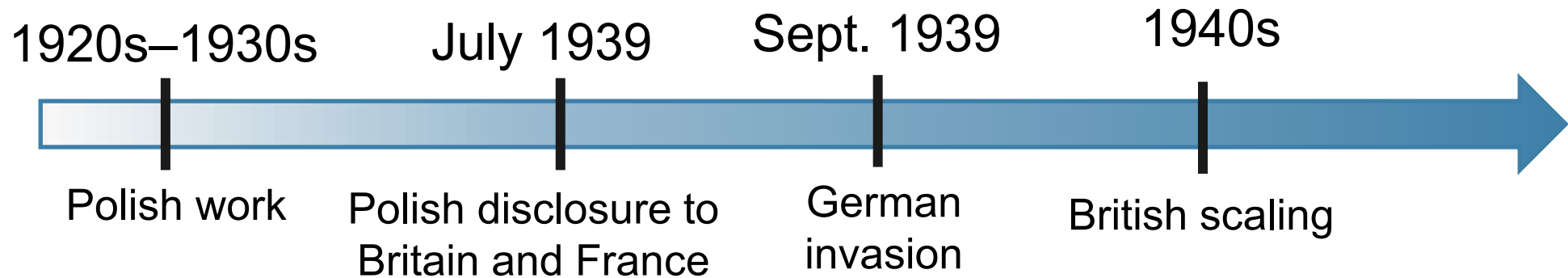
Repeated message key: P G H P G H (after encrypting it with the daily settings)

Ciphertext indicator: K I V B J E

- position **1** and **4** are the **same plaintext letter**: P;
- position **2** and position **5** are the **same plaintext letter**: G;
- position **3** and position **6** are the **same plaintext letter**: H.

3. **Mathematical modelling:** Rejewski used the repeated structure to build relationships between letters.
4. **Mechanized search:** The Polish built catalogues and early “bomba” machines to find daily settings faster.

From Poland to Bletchley: a breakthrough transferred before the war



The British did not start from zero. **Polish contribution:**

- Rejewski modelled Enigma mathematically.
- Polish analysts exploited German message-key procedures.
- They reconstructed machine behaviour.
- They built early mechanized aids: the Polish **bomba**.
- In July 1939, Poland shared its work with Britain and France.

Strategic significance

Poland transformed Enigma from an “impossible” problem into an attackable system.



Bletchley Park: from cryptanalysis to intelligence production

Bletchley Park turned cryptanalysis into organized production.

It required:

- radio interception networks;
- mathematicians and cryptanalysts;
- engineers and machine operators;
- linguists and translators;
- traffic and intelligence analysts;
- usable intelligence, known under **Ultra** classification;
- secure dissemination to commanders.

Y-stations
radio interception



Bletchley Park
cryptanalysis and machine search



Translation and intelligence
analysis



Ultra reports



Commanders and decision -
makers



Breaking Enigma was not a single event.
It was a continuous national defence process.



Bletchley Park, 1939–1946

Chosen as Britain's secret codebreaking centre just before World War II, Bletchley Park became home to “Government Code and Cypher School” GC&CS. During 1940–42, Bletchley changed from a small secret unit into an industrial-scale intelligence operation. Intercepted enemy wireless traffic came from “Y stations,” was processed in huts and blocks around the estate, and became **Ultra** intelligence.

By 1943 - 44, Bletchley had expanded massively.

Staff peaked at **over 9,000**, many of them women, and the work moved beyond Enigma to systems such as the German **Lorenz** cipher. The need to attack Lorenz led to **Colossus**, delivered to Bletchley in early 1944, one of the landmark machines in the birth of modern electronic computing. At the end of the war in 1945, much of the workforce was released, and staff remained bound by secrecy. [GCHQ](#) records note that some wartime staff were told, after hostilities ceased, that there was “no further work” for them in the organisation.

Bletchley Park's life as the main codebreaking centre effectively ended in **1946**, when the post-war signals-intelligence organisation moved to **Eastcote**; GC&CS evolved into what became **GCHQ**. The site later had various government and educational uses, declined badly, and was eventually₄₃ preserved as the Bletchley Park museum.



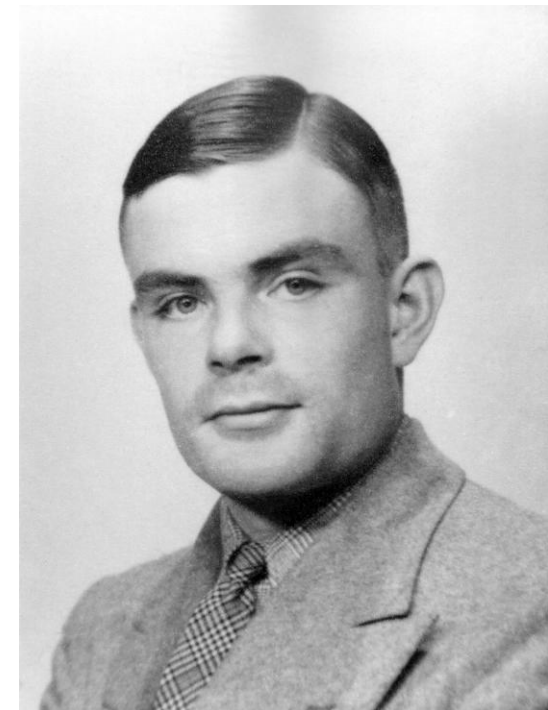
Turing's problem: reduce the impossible search

The problem: Trying every Enigma setting directly was infeasible.

The insight: Do not search blindly.

Use constraints. Turing's approach exploited:

- guessed plaintext fragments, called **cribs**, i.e., guessed fragments of plaintext
 - for example, if a message is a weather report, it may contain predictable words
 - if a German report has a standard format, certain words may appear in certain positions.
- known machine properties;
- logical relations between plaintext and ciphertext;
- automated elimination of impossible settings.



Alan Turing

Goal: Find candidate daily settings fast enough for intelligence to be useful.



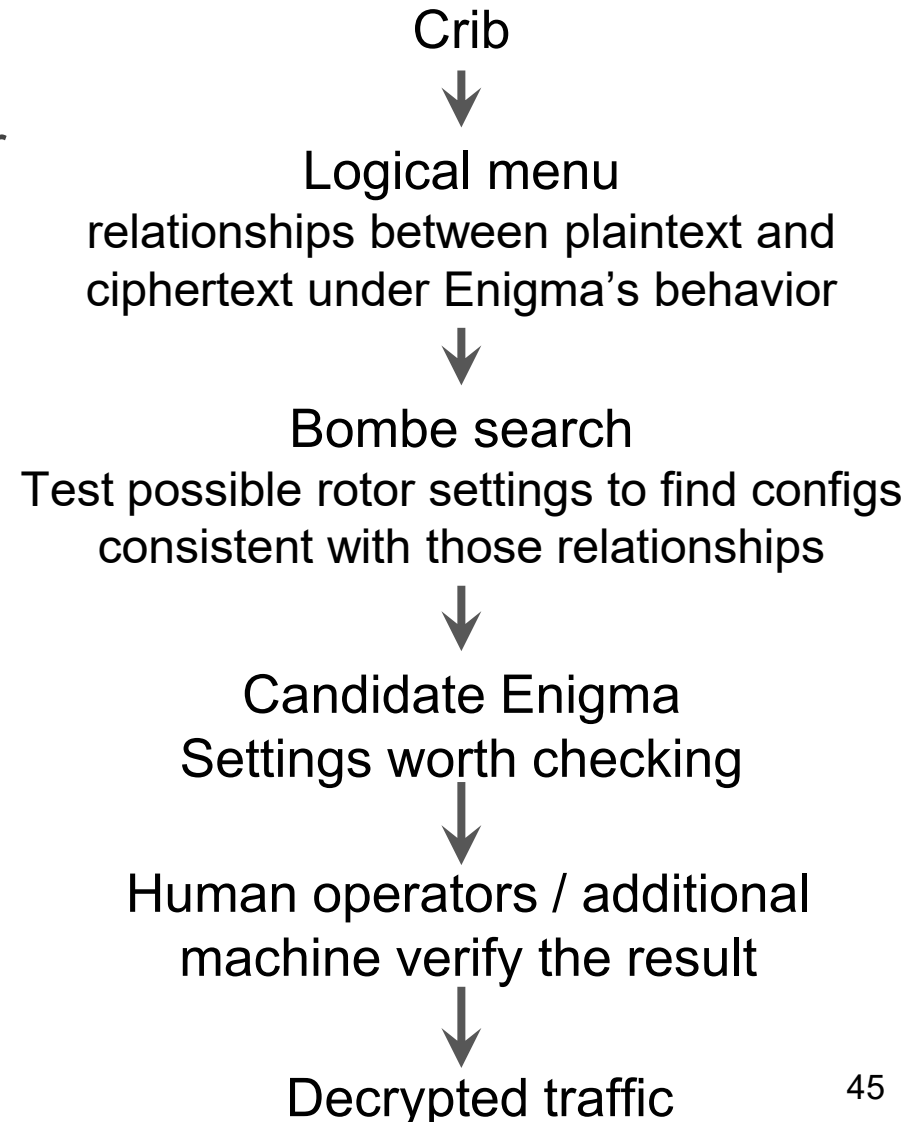
The Bombe: making cryptanalysis operationally fast

The British Bombe automated the search for possible Enigma settings. It did not “magically decrypt” messages. It tested logical constraints derived from cribs and machine structure.

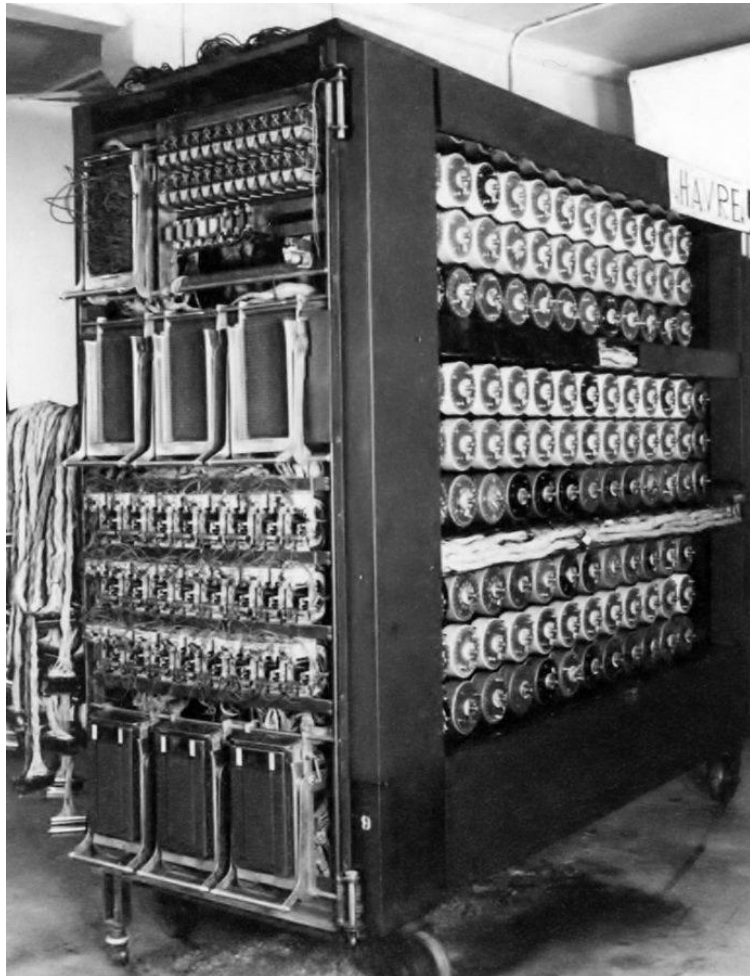
Why it mattered

- reduced search time;
- produced candidate daily settings;
- made decryption operationally useful;
- turned cryptanalysis into repeatable production.

Speed mattered: intelligence had to arrive before it expired.



Alan Turing Bombe





Ultra: when decrypted traffic becomes strategic intelligence

Ultra was the strategic intelligence advantage created from decrypted enemy communications

- German commanders believed Enigma protected their military communications.
- Allied intelligence could read selected parts of that communication system and turn them into operational advantage.

Where it mattered

- **Battle of the Atlantic:** U-boat activity and convoy protection
- **North Africa and Mediterranean:** supply lines and operational planning
- **D-Day:** German deployments and deception assessment

Ultra was powerful, but not omniscient. Not every message was read. Not every message was read in time. And intelligence still had to be interpreted correctly.

✦ The strategic asset was the **information asymmetry**: one side trusted the system, the other side secretly exploited it.



If you act too perfectly, the adversary learns

Using intelligence can reveal that you have intelligence.

Ultra created an operational dilemma:

- **Direct action:** tactical advantage / risk exposing the source if act too visibly
- **Covered action:** plausible explanation / source preserved
- **No action:** tactical cost / long-term access preserved

If every Allied action was too accurate, Germany could infer that Enigma was compromised. Ultra often had to be used indirectly:

- sent reconnaissance aircraft to simulate visual observation;
- false sightings;
- attack only some targets, not all of them;
- sometimes delayed or avoided action.



The source may be more valuable than a single target.

Machines are strong; procedures are fragile

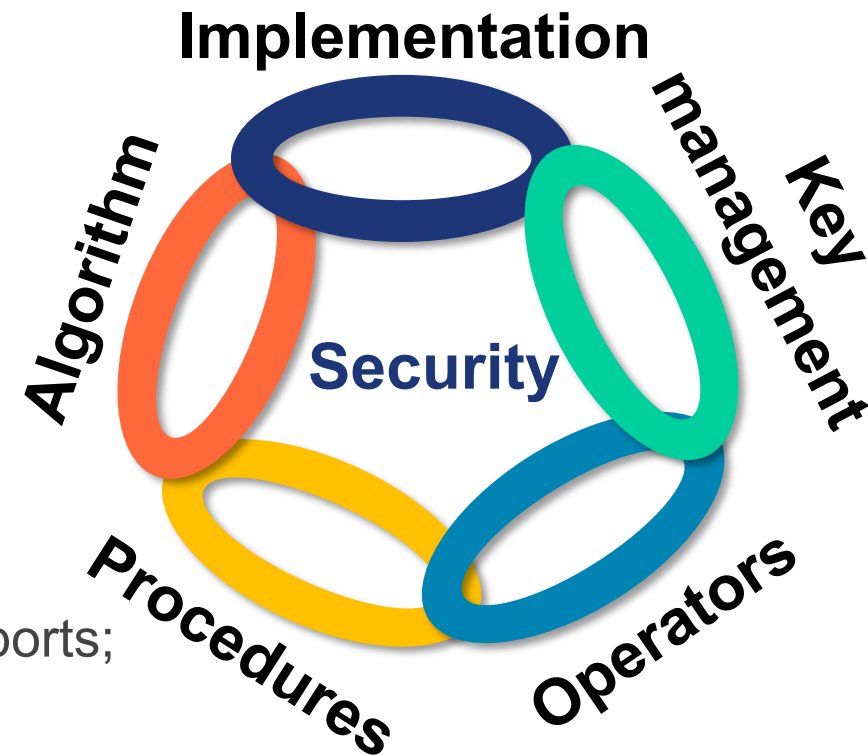
Enigma did not fail only as a machine. It failed as a system.

Technical strength:

- huge key space;
- rotors and plugboard;
- daily keys and message keys;
- dynamic encryption.

Operational weaknesses:

- repeated message-key **procedures**;
- predictable **operator** choices, called *cillies*;
- stereotyped messages, such as weather reports;
- captured **key** sheets and codebooks;
- rules that accidentally reduced randomness.



✦ A strong algorithm cannot compensate for weak operational discipline.



After the war: Ultra remained a state secret

A broken cryptographic system can remain an intelligence asset after the war.

After 1945:

- Bletchley Park's success against Enigma remained secret.
- Thousands of captured Enigma machines were distributed to former British colonies and Commonwealth countries.
- *These countries believed Enigma was secure*; Britain did not correct that belief.
- For years, British intelligence could continue reading some confidential traffic.

The human cost of secrecy:

- Bletchley Park was closed. The Bombes were dismantled.
- Documents were destroyed or locked away.
- Many cryptanalysts returned to civilian life.
- They were bound by secrecy and could not explain their wartime contribution. Key Example: Alan Turing. Turing's work was central to breaking Enigma at scale. His wartime contribution remained classified. He died before public recognition of Bletchley's role became possible.

DES and the Politics of Cryptographic Standards

When cryptography moved from secret machines to public infrastructure



From wartime secrecy to digital infrastructure

After World War II, cryptography moved from secret machines to networked computers.

- Wartime cryptanalysis accelerated the development of computing.
- Computers became faster, cheaper and more widely available.
- In the 1950s and 1960s, **business computing** started to grow.
- Companies began to store and process sensitive digital information: financial transactions, business records, contracts and sensitive data.
- Early **networks**, such as **ARPANET**, connected remote computers.
- **Secure communication became a civilian and commercial need.**



How do we protect sensitive data when computers and networks become shared infrastructure?



Why a public encryption standard was needed

Civilian cryptography needed interoperability. If every organization used a different encryption system:

- encrypted data could not easily be exchanged;
- products would not interoperate;
- banks and companies would need custom solutions;
- trust would depend on private, incompatible designs.

The U.S. National Bureau of Standards looked for a common encryption standard.

IBM proposed a cipher derived from **Lucifer**, developed by Horst Feistel.

Lucifer became the basis for **Data Encryption Standard (DES)**.



**When cryptography becomes infrastructure,
standardization becomes strategic.**



DES: a public standard under classified influence

DES was published as the federal standard FIPS PUB 46 on January 15, 1977. But its design reflected national-security intervention.

DES was based on IBM's Lucifer, but the final standard was controversial. Why?

- The U.S. National Security Agency (NSA) was involved in the evaluation and modification process.
- **The key length was limited to 56 bits.**
- Public researchers worried that **this made brute-force attack feasible for powerful actors**, like state-level actors with advanced computing resources.
- The internal design choices were not fully explained publicly.
- This created suspicion about whether the algorithm had been weakened.

NSA influence created a public trust problem because some design decisions were classified or opaque.



A public cryptographic standard can still depend on classified choices.



DES did not solve the key-distribution problem

DES was symmetric: Alice and Bob still needed the same secret key.

DES helped solve:

- interoperability through a common algorithm;
- commercial use of encryption;
- protection of stored and transmitted data.

But DES did not solve:

- how Alice and Bob securely obtain the same secret key;
- how banks distribute keys to thousands of customers;
- how organizations rotate and revoke keys;
- how large networks avoid courier-based key logistics.



The algorithm was standardized, but the key still had to travel securely.



Key distribution becomes a strategic bottleneck

Traditional model

Secure communication required:

- secret keys;
- secure delivery;
- trusted couriers;
- key storage;
- key rotation;
- destruction of expired keys.

This was manageable for states, but difficult for civilian networks.

Governments could invest in large communication-security logistics.

Commercial and public networks could not scale the same model easily.

Networked world problem

More users + more remote communication = more keys to distribute.

- 5 users → 10 key relationships
- 1000 users → key-management explosion

Can two parties establish secure communication without first pre-sharing a secret?

This question became urgent because of networks.

Case 4: GCHQ and public-key cryptography

When the public revolution had a
classified prehistory

The revolutionary question

Can secrecy be established over a public channel?

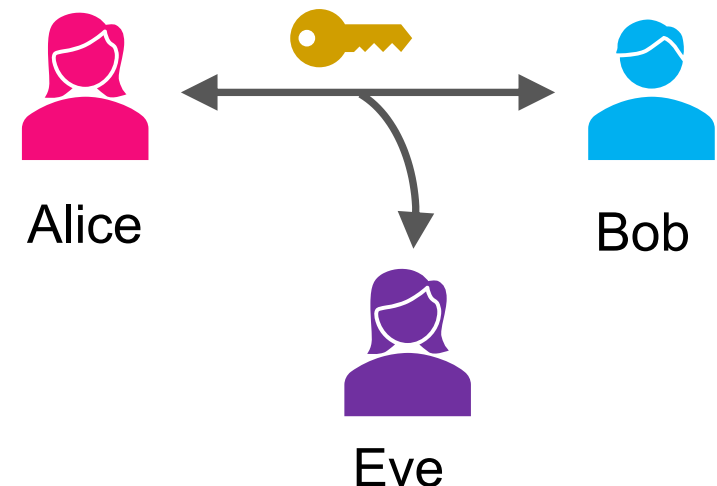
Traditional symmetric cryptography:

- Alice and Bob use the same secret key.
- The key must be shared before communication.
- If Eve intercepts the key, the system fails.

Can Alice and Bob communicate securely without first sharing a secret?

Why this mattered:

- computer networks were growing;
- remote users needed secure communication;
- courier-based key distribution could not scale;
- civilian cryptography needed **a new model**.



The public breakthrough: Diffie-Hellman-Merkle

A shared secret from public information

Diffie, Hellman and Merkle showed that:

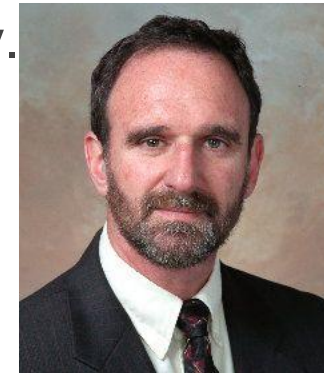
- Alice and Bob can exchange public information.
- Each keeps private information secret.
- Both derive the same shared key.
- Eve sees the public exchange, but cannot derive the key efficiently.

What this solved

- no pre-shared secret required;
- no courier required;
- suitable for open networks;
- **key distribution becomes a protocol, not a logistics operation**



Withfield Diffie



Martin Hellman



The public channel becomes part of the security mechanism.



Public-key encryption: split the key

One key can be public; the other must remain private.

Symmetric model

- same key encrypts and decrypts
- the key must stay secret
- needs key distribution

Public-key model

- Public key can be distributed to everyone.
- Bob, or anyone else can use Alice's public key to encrypt a message for her. But the public key cannot decrypt the message.
- Only Alice's private key can do that.

RSA: making public-key encryption from an idea to a practical system

A mathematical trapdoor made the idea usable.

RSA was developed by:

- Ronald Rivest (**R**)
- Adi Shamir (**S**)
- Leonard Adleman (**A**)

Core intuition

- Multiplying two large prime numbers is easy.
- Factoring the product back into the primes is hard.
- The product can be public.
- The original primes remain private.



Rivest

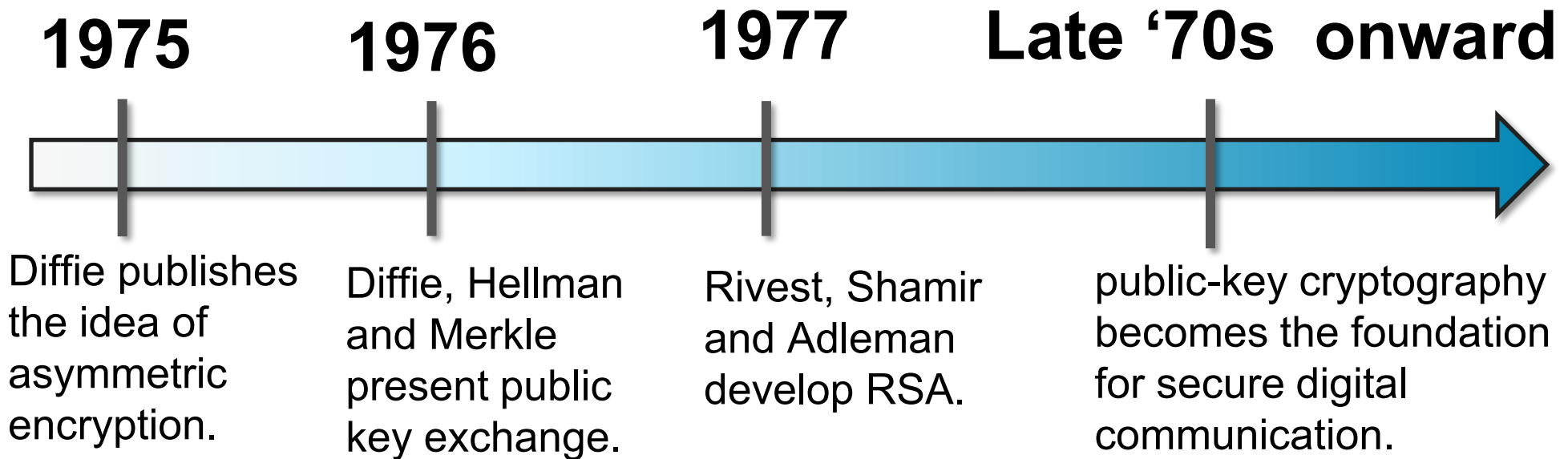
Shamir

Adleman

Result: Anyone can encrypt using public information. Only the holder of the private information can decrypt.

The public history everybody knew

Stanford and MIT changed modern cryptography.



Public impact:

- secure key establishment;
- encrypted communication over open networks;
- foundations for online banking, e-commerce, secure web protocols and digital signatures.



The classified prehistory: GCHQ

The same revolution had already begun inside British intelligence (GCHQ):

- late 1960s: British defence planners worried about key distribution, in a world in which radio communication would become cheaper, smaller and more widespread; If every soldier or unit needed secure communication, then distributing secret keys to everyone would become almost impossible;
- **James Ellis** proposed the idea of “non-secret encryption”;
- **Clifford Cocks** developed a mathematical RSA-like method in **1973**;
- **Malcolm Williamson** developed a Diffie-Hellman-like key exchange in **1974**;
- the work remained classified.

The public-key revolution happened twice: **once in secret, once in public.**

✦ The classified domain may be ahead of the public domain, sometimes by years.



James Ellis: the idea of non-secret encryption

James Ellis asked a radical question: Do we really need to send the secret key at all?

His contribution:

- identified the key-distribution problem as the central obstacle;
- imagined “non-secret encryption”;
- separated the idea of encryption from the physical delivery of a secret key;
- showed that public-key cryptography was conceptually possible.

Limitation

Ellis had the idea, but not yet the mathematical function needed to implement it.

📌 Strategic innovation often begins by challenging the assumption everyone else accepts.



Cocks and Williamson: the missing mathematics

GCHQ found both pieces of the public-key revolution.

Clifford Cocks, 1973

- joined GCHQ after Cambridge;
- used number theory;
- developed an RSA-like public-key encryption method;
- discovered the idea years before RSA became public.

Malcolm Williamson, 1974

- tried to find a flaw in Cocks's idea;
- instead developed a key-exchange method;
- independently reached a Diffie-Hellman-like solution.



Pure mathematics can become a
national-security capability.



Why the discovery remained secret

In intelligence, discovery does not automatically become publication.

Why secrecy made sense to GCHQ

- cryptography was linked to military and diplomatic communications;
- revealing the work could reveal state capability;
- publication was incompatible with intelligence culture;
- patenting would have required disclosure;
- public-key methods were computationally heavy for early-1970s machines.

But secrecy had costs

- no publication, no public recognition, no patent advantage;
- no early civilian diffusion;
- the public story developed without GCHQ.



Classified advantage can become strategic opportunity ... or strategic missed opportunity.

Public science vs classified intelligence

Knowledge flows differently in open and secret communities.

Public science

- publishes discoveries;
- rewards priority and citation;
- enables verification and adoption;
- accelerates innovation.

Classified intelligence

- restricts disclosure;
- protects capabilities;
- limits adversary learning;
- may hide discoveries for decades.

Asymmetry

The classified world
can read public
research.
The public world
cannot read
classified research.





Recognition delayed: the 1997 disclosure

The classified history became public almost three decades later.

What happened

- GCHQ considered disclosure in the 1980s.
- Disclosure was delayed.
- In 1997 Clifford Cocks publicly described the GCHQ contribution.
- Ellis, Cocks and Williamson finally received recognition.
- James Ellis died shortly before the public recognition.

Historical pattern

- Babbage and Vigenère cryptanalysis;
- Turing and Enigma;
- Ellis, Cocks and Williamson and public-key cryptography.



Cryptographic knowledge is itself a strategic asset

Cryptographic knowledge is itself a strategic asset.

These cases teaches that states can exercise cryptographic power by controlling:

- research agendas;
- classification;
- disclosure timing;
- standardization influence;
- patents and economic exploitation;
- recognition and historical narrative.

A state may be ahead in cryptography even when the public world does not know it.

Case 5: PGP and the Crypto Wars

What happens when strong cryptography
leaves the state?



The new battlefield: strong cryptography leaves the state

PGP changed the political meaning of cryptography.

Before the digital age

- Strong cryptography was mainly a state, military and diplomatic capability.
- Secure systems were expensive, specialized and organizationally controlled.
- Access to advanced cryptography was limited.

With personal computers and the Internet

- strong cryptography could run on ordinary machines;
- software could be copied globally;
- users could encrypt without asking the state;
- encryption became a mass technology.

Should everyone be allowed to use strong cryptography?



The Crypto Wars began when cryptographic power became decentralized.

Phil Zimmermann: bringing strong cryptography to ordinary users

A software engineer turned public-key cryptography into a tool for citizens.

Who was Phil Zimmermann?

- An American software engineer and privacy advocate.
- He believed that digital communication needed strong privacy protections.
- He saw public-key cryptography as too important to remain only in institutional or expert hands.
- In 1991, he released **Pretty Good Privacy (PGP)**.
- PGP made strong encryption usable for ordinary e-mail users.



Phil Zimmermann

Why Zimmermann cared about digital privacy

Digital communication made surveillance easier to automate.

Zimmermann's concern: Digital communication was convenient, but also easier to monitor at scale.

Traditional communication

- paper letters were harder to search automatically;
- interception was costly and selective;
- surveillance required people, access and time.

Digital communication

- e-mails pass through servers;
- messages can be copied silently;
- filtering can be automated;
- **surveillance can scale.**

For Zimmermann, PGP was not only software. It was a defence against scalable surveillance.

Zimmermann answer



Use strong
cryptography to
give ordinary
users privacy in
the digital world.



Zimmermann's idea: PGP as usable cryptography

PGP packaged strong cryptography into software people could actually use.

PGP = Pretty Good Privacy

Zimmermann combined:

- public-key cryptography;
- symmetric encryption;
- digital signatures;
- key management;
- an e-mail-oriented workflow.

Why this mattered

- no special hardware; no state infrastructure; no institutional permission; software could be copied globally.

PGP did not
invent the theory.
It made the theory
usable and
distributable.



How PGP worked: hybrid encryption

PGP combined speed and scalable key distribution.

Problem: RSA was powerful but slow for long messages.

PGP solution

1. Generate a random session key for that message.
2. Encrypt the message with a fast symmetric cipher using the session key.
3. Encrypt the session key with the recipient's public RSA key.
4. Send both: encrypted message; encrypted session key.

Why it mattered

- fast enough for personal computers;
- secure enough for strong privacy

PGP also provided digital signatures

Confidentiality was not enough: users also needed authenticity.
Digital communication creates a second problem: **How does Bob know the message really came from Alice?**

PGP supported digital signatures

- Alice signs using her private key.
- Bob verifies using Alice's public key.
- The signature helps prove origin and integrity.

Security goals

- **Confidentiality:** only Bob can read;
- **Integrity:** Bob knows it was not modified;
- **Authenticity:** Bob knows Alice sent it.



PGP protected
both
confidentiality and
trust.



PGP goes online: software becomes uncontrollable

Once released on the Internet, strong cryptography became hard to contain.

1991

- Zimmermann released PGP through online distribution.
- Users could download, copy and redistribute the software.
- PGP spread beyond the United States.
- Human-rights groups, activists and ordinary users adopted it.

Strategic effect

The state could regulate companies and hardware, but software on the Internet was much harder to control.



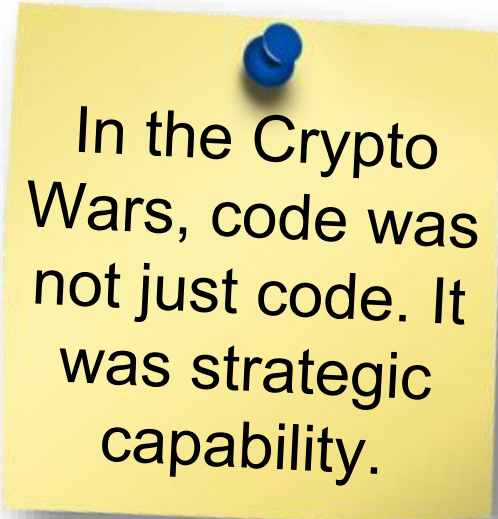
Why the U.S. government reacted: export control

Strong cryptography was treated like a strategic technology. U.S. policy treated strong cryptographic software as export-controlled.

For the government, uncontrolled spread of PGP could:

- reduce intelligence collection;
- protect foreign adversaries;
- help criminal or terrorist networks;
- weaken lawful interception;
- reduce state visibility over communications.

Zimmermann was investigated because PGP had spread outside the U.S.: PGP was software, but the state treated its export as a national-security issue.



In the Crypto Wars, code was not just code. It was strategic capability.



Zimmermann investigated, then PGP became legitimate

Trying to suppress PGP increased its symbolic power.

Timeline:

- 1991: PGP released online.
- 1993: Zimmermann investigated for alleged illegal export.
- 1996: investigation ends without indictment.
- Later: PGP becomes a legitimate privacy and commercial tool.

Why the investigation mattered:

- software, speech, publication and export overlapped;
- civil-liberties groups rallied around the case;
- PGP became more famous;
- legal control became harder once software was globally distributed.



Once strong cryptography is globally distributed, suppression may accelerate diffusion.



Export control did not stay in law books

Legal limits became technical limits. Export controls affected three layers:

1. **Distribution:** Who may receive strong cryptography?
2. **Products:** What strength may vendors ship internationally?
3. **Architecture:** What cryptographic functions are exposed by software, hardware and standards?

Two concrete examples:

- **Browsers:** export-grade encryption for foreign users.
- **TPM:** hardware/standard design choices constrained what cryptographic functions were exposed.



Cryptographic policy can become product design.

Use case A: export-grade browsers

The same Internet did not mean the same security.

Export-grade browser cryptography:

- U.S. domestic users could receive stronger encryption.
- Foreign users often received weaker export-grade encryption.
- International e-commerce therefore depended on weaker protection.
- This created trust and competitiveness problems.
- Weak legacy modes could remain in protocols and products even after policy changed.



Bob in the U.S.
Stronger encryption



Alice abroad
Export-grade encryption



Cybersec Export policy created different security classes for different users.



Use case B: TPM and export-sensitive architecture

A hidden example: policy constraints can shape hardware boundaries.

TPM 1.2	TPM 2.0
centered mainly on RSA and SHA-1	introduced algorithm agility
did not expose symmetric operations for general message encryption, or bulk symmetric encryption capabilities	supports symmetric cryptography, including AES
reflected cost and export-control constraints	still serves mainly as a root of trust, not as a high-speed general-purpose data-encryption accelerator



The boundary of a cryptographic product is also a policy and sovereignty question.



Why export controls became hard to sustain

The Internet broke the old border-control model.

Export restrictions became unstable because:

- software could be copied globally;
- source code could be published, printed and mirrored;
- strong cryptography became available outside the United States;
- U.S. companies needed global competitiveness;
- e-commerce required strong cryptography;
- civil-liberties groups challenged restrictions on code and publication.

Policy shift:

- 1991: PGP goes online;
- 1990s: export-grade cryptography problem;
- 1996: commercial encryption control shifts toward Commerce Department management;
- 2000: major liberalization for many commercial encryption products;
- ... **but encryption controls do not disappear completely.**



The battlefield changes: from export control to access control

If strong cryptography cannot be stopped, can access still be preserved?

Export control question: Can we stop strong crypto from spreading?

New access question: If it spreads, can we still access plaintext?

State concerns:

- lawful interception;
- criminal investigations;
- counterterrorism;
- counterintelligence;
- signals intelligence;
- strategic visibility.

Civilian and business concerns:

- privacy;
- protection of journalist sources;
- secure commerce;
- freedom of expression;
- protection from abuse;
- trust in digital infrastructure.

✦ The Crypto Wars shift from controlling distribution to controlling access.

Key escrow: the proposed compromise

Strong encryption, but with recoverable access.

Key escrow idea: Users may use strong encryption, but recovery access exists.



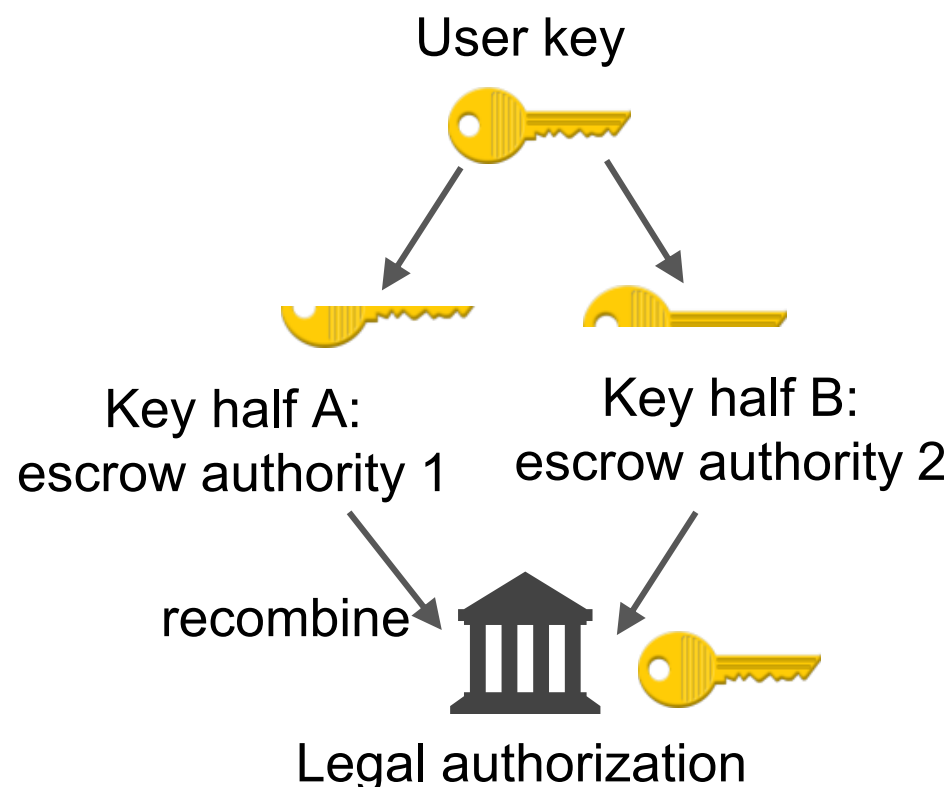
Goal:

- users get strong cryptography;
- law enforcement can recover keys with legal authorization;
- the state preserves exceptional access.

Core trust problem:

Who holds the recovery capability?

Risks: abuse by authorities;
insider compromise; systemic vulnerability;
loss of public trust; international distrust.



Clipper and Capstone: state-designed exceptional access

The U.S. government tried to standardize escrowed encryption. American Escrowed Encryption Standard, 1994:

- Clipper: telephone communications;
- Capstone: computer communications;
- encryption implemented in chips;
- key material split and escrowed;
- access possible under legal conditions.



Clipper –chip photo

Policy objective:

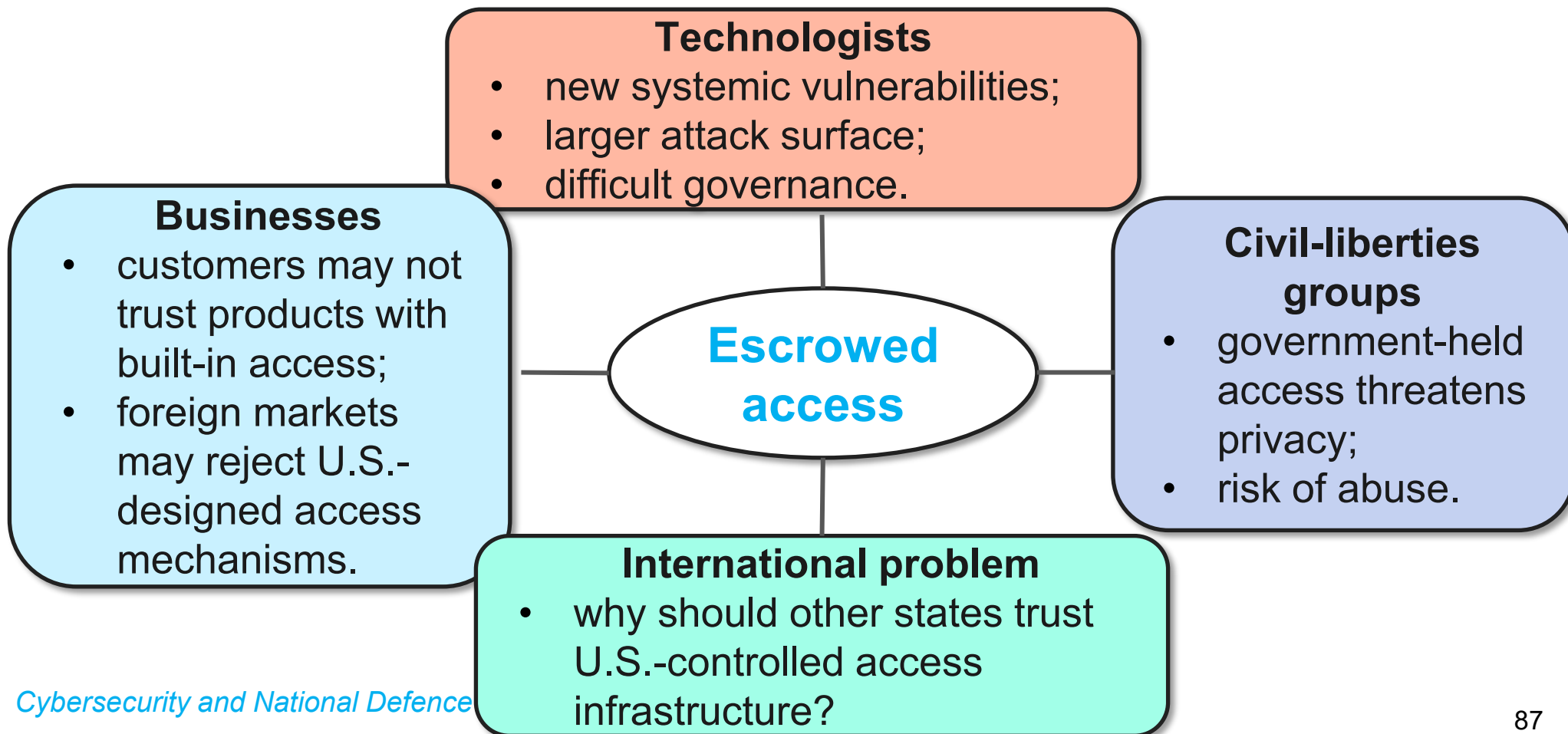
Strong encryption without losing lawful access.

Engineering and civil-liberties objection: Exceptional access becomes part of the attack surface. It must never be abused.

📌 A security standard cannot succeed if the trust model is politically unacceptable.

Why escrow failed: the trust model collapsed

Why key escrow was rejected?





Clipper died, but exceptional access returns

The standard failed, but the political demand did not disappear.

What happened to Clipper / EES?

- 1994: Escrowed Encryption Standard, FIPS 185.
- mid-1990s: Clipper/key escrow failed to gain trust and adoption.
- 2015: FIPS 185 formally withdrawn as obsolete.

But the idea returned in new forms: Modern debates often ask similar questions:

- Can a company be compelled to help unlock a device?
- Can cloud encryption be designed to allow lawful access?
- Can encrypted services be required to detect illegal content?
- Can scanning happen before encryption, on the client side?



Exceptional access was not abandoned; it was renamed, redesigned and repeatedly reintroduced.



Apple 2016

February 16, 2016

A Message to Our Customers

The United States government has demanded that Apple take an unprecedented step which threatens the security of our customers. We oppose this order, which has implications far beyond the legal case at hand.

This moment calls for public discussion, and we want our customers and people around the country to understand what is at stake.

[Answers to your questions about privacy and security >](#)

The Need for Encryption

Smartphones, led by iPhone, have become an essential part of our lives. People use them to store an incredible amount of personal information, from our private conversations to our photos, our music, our notes, our calendars and contacts, our financial information and health data, even where we have been and where we are going.



Apple/iCloud 2025

Apple appeals to overturn UK government's 'back door' order, FT reports

By Reuters

March 4, 2025 8:16 PM GMT+1 · Updated March 4, 2025



March 4 (Reuters) - Apple ([AAPL.O](#))  has appealed a British government order to create a "back door" in its most secure cloud storage systems, the Financial Times reported on Tuesday, citing people familiar with the matter.

The iPhone maker removed its most advanced security encryption for cloud data, called Advanced Data Protection (ADP), in Britain last month, in what was an unprecedented response to government demands for access to user data.



European Parliamentary Research Service

Table 20: Identified solutions for detecting CSAM in E2EE

	Type of content	Effectiveness	Feasibility	Privacy	Security	Transparency
Device related solutions						
All detection done on-device	Known	***	**	***	*	**
On-device full hashing	Known	****	*****	**	**	***
On-device partial hashing	Known	****	***	**	***	**
On-device use of classifiers	Known New	**	**	**	**	***
Server related solutions						
Secure enclave in ESP server	Known New	****	**	**	**	**
Single third-party matching	Known	****	*	**	**	**
Multiple third-parties matching	Known	****	*	***	***	***
Encryption related						
On-device homomorphic encryption	Known	***	*	**	***	***

Ranking runs from * to *****, whereby * represents a low score and ***** a high score.

Solutions marked in blue are those that are deemed most viable.

Source : [CSA proposal IA](#), p. 290

As can be understood from the table, the experts consulted by the European Commission considered the following three technologies as most promising to be developed further:

- (38) On-device full hashing (with matching at server);
- (39) On-device partial hashing (with matching at server);
- (40) Secure enclaves in Electronic Service Provider (ESP) server.



Beyond Clipper: who controls trust infrastructure?

Cryptography depends not only on algorithms, but also on the infrastructure that makes algorithms usable and trusted.

Public-key cryptography solved one problem:

- Alice can use Bob's public key.
- No pre-shared secret is needed.

But it created governance questions:

- Who certifies that the key belongs to Bob?
- Who controls certificate authorities?
- Who manages recovery mechanisms?
- Who can compel, regulate or pressure these actors?

Trust infrastructure includes: Certificate Authorities (CAs); key recovery systems; hardware roots of trust; standards bodies; cloud identity systems.



Cryptographic sovereignty includes control over the infrastructure that defines digital trust.



AES: rebuilding trust after DES

After DES, the standardization process itself had to become trustworthy.

DES created a trust problem:

- NSA involvement was opaque;
- the 56-bit key length was controversial;
- hidden design choices created suspicion;
- the standard was technically useful but politically contested.

AES changed the process:

- open international competition;
 - public analysis by the cryptographic community;
 - 15 candidate algorithms;
 - Rijndael selected in 2000;
 - AES (Rijndael) standardized as FIPS 197 in 2001.
- Rijndael was officially selected for security, performance, efficiency, implementability and flexibility. But its Belgian origin and the open process helped credibility.



For cryptographic standards, transparency is part of trust.



NIST

Information Technology Laboratory

COMPUTER SECURITY RESOURCE CENTER

PROJECTS

CRYPTOGRAPHIC STANDARDS AND GUIDELINES

ARCHIVED CRYPTO PROJECTS

Cryptographic Standards and Guidelines



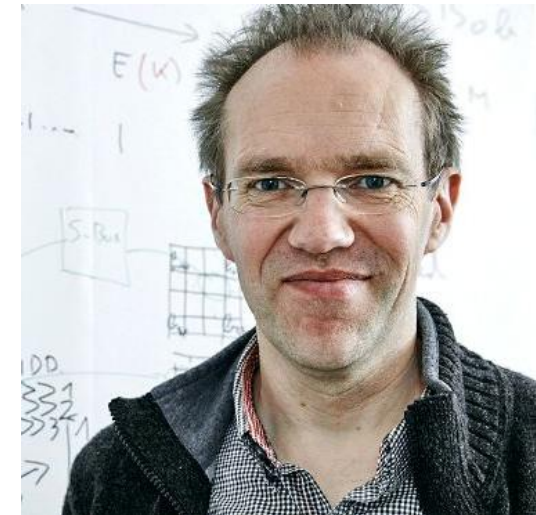
AES Development

[AES Overview](#) | [NIST Reports](#) | [Federal Register Notices](#) | [Rijndael Info](#) | [Related Publications](#)

AES Overview

Beginning in 1997, NIST worked with industry and the cryptographic community to develop an Advanced Encryption Standard (AES). The overall goal was to develop a Federal Information Processing Standard (FIPS) specifying an encryption algorithm capable of protecting sensitive government information well into the 21st century. The algorithm was expected to be used by the U.S. Government and, on a voluntary basis, by the private sector.

On January 2, 1997, NIST announced the [initiation of the AES development effort](#) and received numerous [comments](#). NIST then made a formal [call for algorithms](#) on September 12, 1997. The call stipulated that the AES would specify an unclassified, publicly disclosed encryption algorithm(s), available royalty-free, worldwide.



Joan Daemen



Vincent Rijmen



Sovereign alternatives: China's SM standards

Not every state wants to rely on the same cryptographic trust stack.

China has developed its own commercial cryptographic standards, often called **SM** or **ShangMi** (meaning “commercial cryptography”) algorithms:

- **SM2**: elliptic-curve public-key cryptography;
- **SM3**: cryptographic hash function;
- **SM4**: block cipher for symmetric encryption;
- **ZUC**: stream cipher, used in mobile-communication contexts;
- **SM9**: identity-based cryptography.

Why develop national standards? Reduce dependence on foreign algorithms; support domestic certification and regulation; protect national information infrastructure; strengthen industrial and technological sovereignty; avoid blind trust in foreign standardization ecosystems. China does not live outside global cryptography. It uses, adapts and participates in international standards, but also promotes domestic algorithms for regulated and sovereignty-sensitive contexts.



Final lesson: cryptography is a national-defence layer

Cryptographic sovereignty means controlling what can be **trusted**. States shape cryptography through:

- export controls;
- product capabilities;
- exceptional-access proposals;
- hardware and APIs;
- certificate authorities;
- key recovery systems;
- standards and certification.

Cryptographic sovereignty means the ability to choose, verify, implement, certify and replace standards and products that protect national infrastructure.



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